Problem Statement & Research Agenda

Why this project exists — what's broken, what we're solving, and what we set out to prove

NSE Equities | SEBI Compliant | Open Source | 3-Class Prediction | Walk-Forward Validation

1 The Problem — What Is Broken in Existing Research

The Indian National Stock Exchange (NSE) is one of the world's largest exchanges by volume. Researchers and developers have built hundreds of machine learning models claiming to predict stock direction. **Nearly all of them are misleading** — not because the models are bad, but because they are tested incorrectly and evaluated without accounting for the reality of trading in India.

Three specific failures exist in virtually every prior NSE ML paper:

! Problem 1 — Data Leakage in Testing

Most projects use a simple 80/20 train-test split. Because features like "5-day return" span the split boundary, the model accidentally uses future data to predict the past. Results look good in testing but collapse in production.

! Problem 2 — Transaction Costs Completely Ignored

Every trade on the NSE incurs mandatory statutory fees: Securities Transaction Tax (STT 0.10%), brokerage, GST, stamp duty, and SEBI turnover charges. These total ~0.34% per trade — ~0.64% round-trip. Papers reporting "profitable strategies" almost universally ignore this. A model that looks great on paper loses money in reality.

! Problem 3 — Binary Classification Ignores the FLAT Reality

Most models predict only UP or DOWN. In reality, 40–47% of trading days result in a price change smaller than the cost of trading (FLAT). Forcing binary classification turns these days into random noise that confuses the model and generates fee-burning trades.

Core Truth: A model that claims 85% accuracy on a test set but ignores data leakage and trading costs might actually lose money. The market does not care about test set accuracy.

2 Our Agenda — What Clearward Is Designed to Prove

Clearward is a platform built to answer three specific research questions with complete methodological honesty — using real data, real costs, and leakage-free validation:

Q1 Does adding Hidden Markov Model regime detection + news sentiment improve prediction accuracy?

Finding: HMM regime features add +2.58 percentage points in Macro-F1. Sentiment alone slightly reduces it (−0.53pp) because financial news is reactive, not predictive — but it enables explainability features.

Q2 Does parallelizing walk-forward cross-validation actually reduce wall-clock training time?

Finding: Yes — ThreadPoolExecutor with 4 workers produces a measured 3.37× speedup (from ~25s to ~7.4s per stock), verified by the test suite.

Q3 What does a fee-aware backtester reveal about real profitability on the NSE?

Finding: Gross strategy Sharpe = 0.82 (looks strong). After applying all NSE statutory costs: Net Sharpe = 0.41 — below a passive buy-and-hold strategy (0.55). The model generates real alpha that fees consume at 3-day trading frequency.

The Agenda in One Sentence: Build a machine learning system for NSE direction prediction that is methodologically honest — no leakage, no missing costs, no cherry-picked metrics — so the results can be trusted, replicated, and built upon.

Model Inputs — What the System Receives

Raw data sources, preprocessing pipeline, and all 17 engineered features across 3 tiers

14 Technical Features | 2 HMM Regime Features | 1 Sentiment Feature

3 Raw Data Sources

DATA TYPE	SOURCE	WHAT WE COLLECT	FREQUENCY
Price History (OHLCV)	Yahoo Finance API via `yfinance`	Open, High, Low, Close, Volume — split & dividend adjusted	Daily (3 years: 2022–2024)
Stock Universe	NSE Symbol Catalog	347 NSE/BSE ticker symbols with sector classification	Static + periodic updates
Financial News	RSS Feeds — Economic Times, Moneycontrol, LiveMint	Headlines per stock, MD5-deduplicated, stored in ChromaDB	Continuous scraping
Mutual Fund NAVs	AMFI India via `mfapi.in`	Daily Net Asset Values for 2,500+ schemes	Daily (23-hour cache)
Shareholding Data	NSE Corporate API / Yahoo Finance	Promoter %, FII %, DII %, Retail % ownership	Quarterly filing cycle

4 Preprocessing Pipeline — Raw Data → Model-Ready Input

Yahoo Finance OHLCV Download → SQLite Storage (WAL Mode) → 200-Day Warm-Up Drop → 17-Feature Computation → 3-Day Forward Label: UP/FLAT/DOWN → Walk-Forward Folds (×5)

Labeling Rule: For each day, the 3-day forward return is computed. If > +0.8% → **UP**. If < −0.8% → **DOWN**. Otherwise → **FLAT**. The 0.8% threshold equals the per-leg cost floor — any move smaller than this cannot be profitably traded after fees.

5 The 17 Input Features — 3 Tiers

TIER 1 · TECHNICAL (14)

- `ret_1d` 1-day return — short-term momentum
- `ret_2d` 2-day return
- `ret_3d` 3-day return — matches prediction target
- `ret_5d` 5-day / weekly return
- `ret_10d` 10-day / bi-weekly return
- `rsi_14` RSI — overbought/oversold detector
- `macd_hist` MACD histogram — trend acceleration
- `bb_pct_b` Bollinger %B — range position
- `vol_ratio` Volume vs 20-day average
- `atr_pct` ATR-14 — normalized volatility
- `ema_ratio` EMA-20 / SMA-50 — trend spread
- `stoch_k` 14-period Stochastic %K oscillator
- `price_sma200` Price / SMA-200 — institutional level
- `intraday_range` Candle body conviction (Close−Open)/(H−L)

TIER 2 · HMM REGIME (2)

- `regime_state` Hard Viterbi state: 0=Bull, 1=Bear
- `regime_prob_bear` P(Bear | all past returns) — continuous 0 to 1

Why 2 features?
The continuous probability (ranked **#4 of 17**) captures uncertainty at regime transitions — far more useful than the hard state (ranked #16). A 0.7 bear probability tells the model more than just "bear = yes."

How it's built:
2-state Gaussian Hidden Markov Model implemented from scratch using only NumPy + SciPy. Uses Baum-Welch Expectation-Maximization. No external library. Research basis: Hamilton (1989) and Ang & Timmermann (2012).

TIER 3 · SENTIMENT (1)

- `sentiment` VADER 3-day rolling compound score: −1.0 to +1.0

How VADER works:
VADER (Hutto & Gilbert, 2014) uses a hand-curated lexicon where each word has an emotional valence score. It understands negation ("not good"), capitalization ("CRASH"), and punctuation ("Surge!!!"). Runs in milliseconds on CPU — no GPU required.

Honest finding:
News sentiment *reduces* Macro-F1 by −0.53pp when used for 3-day direction prediction. Financial news is reactive (reports on moves already made), not predictive. It was retained for the Hype Guard and RAG explanation features.

Combined Input Shape: For each stock-day observation, the model receives a single row of **17 numbers** (no raw prices, no images, no sequences). Each number is a derived signal computed from OHLCV history and news text. This row is what the Random Forest reads.

Model Outputs — What the System Delivers

Five distinct outputs, real benchmark numbers, and what they mean in the real world

Direction Prediction | Price Forecast | RAG Explanation | Portfolio Risk | Hype Guard

6 The 5 Outputs the System Produces

Output 1 — Direction Prediction

What: The 150-tree Random Forest votes across UP / FLAT / DOWN and returns a class + confidence score.

SEBI Rule: Labels are legally formatted — never "BUY/SELL." Only: ▲ *POSITIVE BIAS*, ▼ *NEGATIVE BIAS*, — *CONSOLIDATION*.

```
{
  "direction": "POSITIVE BIAS",
  "confidence": 0.71,
  "raw_class": "UP",
  "macro_f1": 0.2515
}
```

Output 2 — SARIMAX Price Forecast

What: Statistical SARIMAX model predicts the actual *price level* (not direction) for the next 5 trading days, with 80% and 95% confidence cones that widen as uncertainty grows.

Use case: Risk planning, stop-loss levels, not a trade signal.

```
{
  "day_3_forecast": 2418.50,
  "ci_80": [2401.2, 2435.8],
  "ci_95": [2389.1, 2447.9],
  "aic_order": "(1,1,1)"
}
```

Output 3 — RAG Plain-English Explanation

What: Retrieves the 3 most relevant news headlines from ChromaDB vector store using semantic search (MiniLM-L6 embeddings). Combines with technical evidence to explain *why* the prediction was made — grounded in real evidence, not hallucinated.

```
"RSI at 71 (overbought) and MACD
decelerating. Supported by: [TCS Q2
revenue beats estimates by 3.2%]
from Moneycontrol (Oct 3, 2024)."
```

Output 4 — Portfolio Risk Audit

What: Given a list of holdings, calculates:

- **Value at Risk (VaR 95%)** — worst expected daily loss
- **Portfolio Overlap %** — are holdings too similar?
- **Fee Drag** — annual drag from expense ratios
- **Stress Test** — simulates 2008 / COVID-level drops

```
{
  "var_95_pct": -2.34,
  "overlap_pct": 67.2,
  "annual_fee_drag": 0.82,
  "stress_loss": -18.4
}
```

Output 5 — Hype Guard Score

What: Detects when a stock is being hyped on social media or news without fundamental support. Compares news sentiment momentum vs price/volume fundamentals. Returns a "Hype Score" 0–100 and a behavioral nudge warning.

Example trigger: Sentiment score = +0.82 (very bullish news) but price is falling and volume is below average → *"Potential retail hype trap detected."*

```
{
  "hype_score": 78,
  "sentiment": +0.82,
  "vol_ratio": 0.73,
  "price_trend": "DOWN",
  "warning": "Hype-price
   divergence detected"
}
```

7 Real Performance Numbers — Across All 5 Stocks

STOCK	SAMPLE SIZE	F1 SCORE	BINARY ACC	GROSS SHARPE	NET SHARPE
RELIANCE.NS	502 days	0.295	72.1%	0.79	**0.48**
TCS.NS	502 days	0.259	70.8%	0.75	0.39
HDFCBANK.NS	502 days	0.237	69.4%	0.71	0.36
INFY.NS	502 days	**0.154**	**65.1%**	0.62	**0.28**
ICICIBANK.NS	502 days	**0.313**	**74.8%**	**0.88**	**0.54**
AGGREGATE (Pooled)	**2,510 days**	**0.2515**	**73.3%**	**0.82**	**0.41**

+75%
BETTER THAN ALWAYS-FLAT BASELINE (F1 0.1434 → 0.2515)

p<0.001
MCNEMAR'S TEST — RESULTS NOT DUE TO LUCK

0.41
NET SHARPE AFTER ALL NSE FEES (BUY-AND-HOLD = 0.55)

Key Real-World Insight: The model generates genuine, statistically proven predictive signal (+75% above baseline, p<0.001). In Bull Market conditions specifically, net Sharpe reaches 0.58 — beating passive buy-and-hold (0.55). The challenge is that Indian statutory trading costs (especially the 0.10% STT per trade) consume ~50% of gross alpha at 3-day trading frequency.